

Nova v2

A multi-model orchestration pattern for conviction and efficiency — one controller, a deterministic rail, read-only workers, a cross-family verifier, one writer, labelled claims. Evidence-graded, red-teamed, with the canonical brief and a worked cost model.

1. What this optimises for, and the one-paragraph answer

You asked for the best multi-LLM system for *conviction* (you can trust what comes back) and *efficiency* (you pay only for what moves the answer), from a brief write-up to a deep dive, with minimal drift. The evidence points to a shape narrower than "multi-agent" usually means:

One frontier controller (Nova) that owns every non-mechanical decision; a deterministic runtime (the Rail) that owns every handoff, every bound, and every mechanical check and can override Nova on bounds; stateless workers one tier below the controller that only read and return; a verifier from a different model family that grades against the original brief; one writer; and a claim-level grounding pass plus a final deliverable gate before anything reaches you. Depth is bought by widening the read phase under a bounded gap loop, never by lengthening chains or parallelising writing. Conviction is earned rung by rung — snapshot-checked evidence first, entailment second, cross-family judgement third — and carried as a typed label whose name says which check earned it. Efficiency comes from static role-to-tier assignment, low reasoning effort by default, a two-stage escalation maximum, and stop rules keyed to evidence novelty.

What it deliberately is not: agents talking to each other, debate, blended mixture-of-agents synthesis, learned per-query routing, persona prompts, or any loop whose termination lives inside a model (§11).

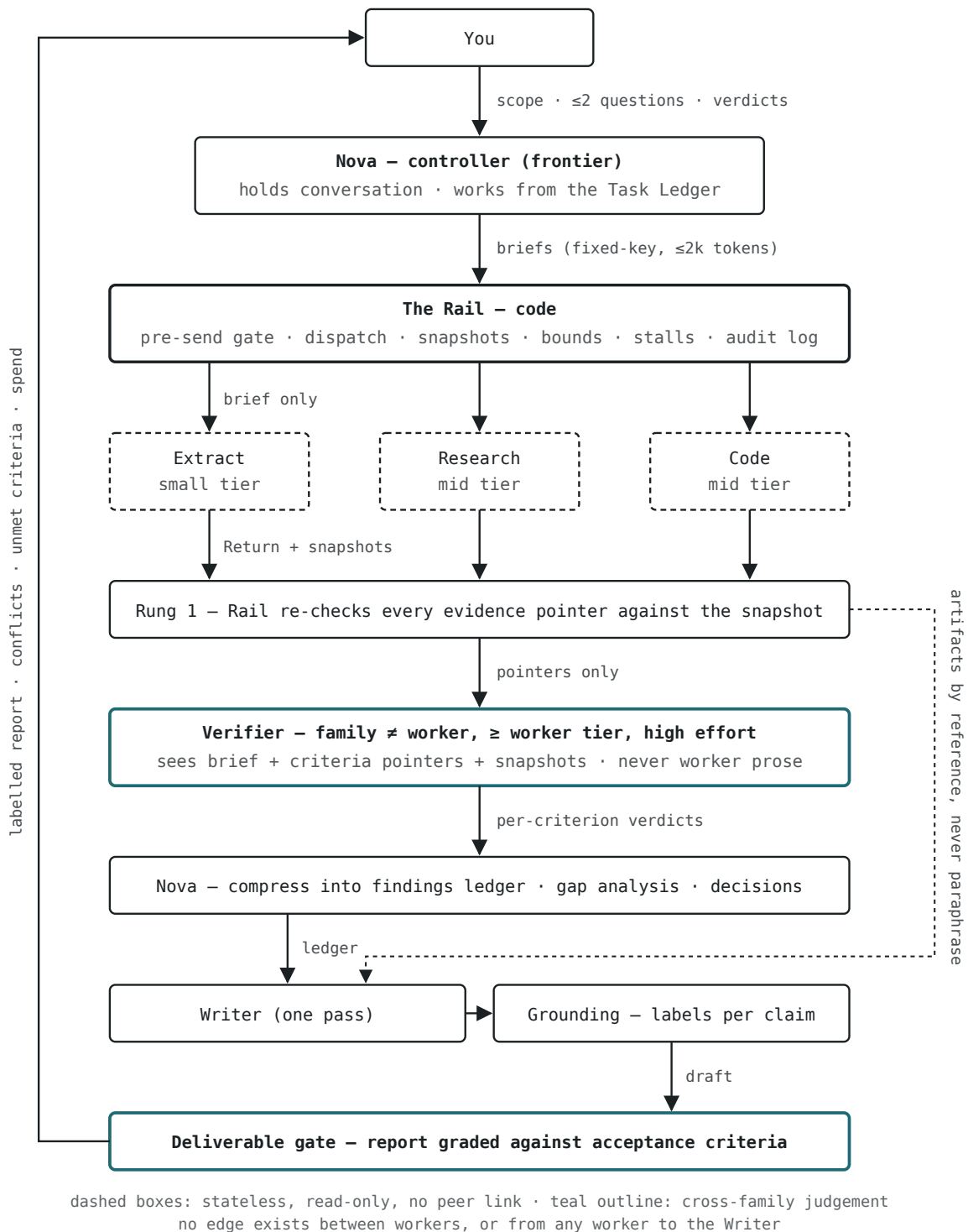
Two things the red-team made me say plainly. First, this design is more expensive per task than a single agent — roughly 2–3× a single-agent run at T2 and around 3× at T3 (§9) — and buys *reliability*, not cheapness; "efficiency" here means no token is spent without a check that can fail. Second, several load-bearing mechanisms (recitation, progress ledgers, reference-not-summary, deterministic rails) are universally shipped and never ablated; they rest on grade-C evidence and §14 says so.

2. The six invariants

Everything else is derived from these.

1. **Nova owns every non-mechanical decision; the Rail owns every bound and every mechanical check, and overrides Nova on bounds.** Centralised coordination contained error amplification to 4.4× versus 17.2× for independent agents in the one multi-topology study ([Google/MIT](#), single preprint); adding a final-authority step and a verification step to ChatDev raised success +9.4% and +15.6% ([MAST](#)). Workers have no write access to shared state, the verifier's inputs, tests, or other workers.
2. **Every handoff is a self-contained, fixed-key, structured brief; no transcript crosses a boundary.** A fully specified single turn recovers ~95% of what incremental delivery loses ([Laban et al.](#) — multi-turn chat, same model; applied here by inference); fixed-key formats lost 4.7 points over six relay hops where prose lost up to 21.8, for weak relay models ([relay study, 2026](#)).
3. **No worker reads another worker's output or the Writer's. All worker output passes through the Rail, the Verifier, and Nova's ledger before the Writer sees it.** The ledger is a relay hop, and the design treats it as one: the Writer reads artifacts by reference for any finding that is not `checked`, never Nova's paraphrase alone (§5, §6). Injected errors persist 83–100% through relays and are never repaired ([relay study](#)).
4. **"Done" is decided by something the worker does not control.** Self-reported completion is false in 13–79% of failures by model ([Confident Closing, 2026](#)); same-family judges favour their own outputs ([Panickssery](#)); intrinsic self-correction is flat or negative ([Huang et al.](#)). Snapshot checks run first, a cross-family Verifier second, Nova last and only for what the first two cannot reach. Stop rules are Rail-owned.
5. **Conviction is a typed label named after the check that earned it, never a scalar and never "verified" by agreement alone.** Verbalised confidence clusters at 80–100% with ECE 0.18–0.52 ([Xiong et al.](#)); frontier-model sample agreement correlates only ρ 0.20–0.59 with correctness and 48% of GPQA answers at agreement ≥ 0.8 were wrong ([2607.08065](#)); scores become semantically incompatible across pipeline boundaries ([2604.23505](#)).
6. **Width over depth; reading over writing; selection over synthesis.** Parallel section-writing produced disjoint reports ([LangChain ODR](#)); diverse candidates help when a judge *selects* (0.81 win rate) and hurt when *blended* (0.18) ([selection bottleneck, 2026](#)).

3. Topology and roles



Every piece of worker output crosses the Rail, the Verifier and Nova's ledger before the Writer sees it; the only route around the ledger's paraphrase is the dashed artifact reference. The two teal boxes are the points where a different model family judges.

ROLE	TIER (DEFAULT)	EFFORT	MAY READ	MAY WRITE	NOTES
Nova	Frontier (Opus-class)	Medium	Ledger; Verifier verdicts; labelled findings; artifacts by ref	Ledger, Briefs, decisions	Never reads raw worker prose before Rail + Verifier have labelled it. Its transcript is compacted by the Rail's policy (\$7.5), and it works from the Ledger, not memory.
The Rail	Code	–	Everything	Dispatch, snapshots, bounds, labels (mechanical ones), audit log	Every loop – including the Nova-gate bounce loop – has a Rail- side bound. 68 confirmed infinite-loop defects across 47 real agent projects lived in framework feedback paths, not top- level counters (Ho u et al.).
Extraction worker	Small (Haiku / Flash-Lite / mini class)	Low	Its brief	Return	Schema + 2 examples in brief. Run 2x, compare at field level (\$8 Rung 2). Escalate to mid tier by Rail rule: input > ~30 pages or schema > ~100 fields – Gemini 3.5 Flash fell from 87.9% on short docs to 27.9% on long (Ext ractBench).
Research worker	Mid (Sonnet / Flash class)	Low	Its brief; web/corpus via Rail-proxied fetch	Return; artifacts to snapshot store	Higher effort was flat or negative on research tasks for 3 of 4 frontier models at 1.5–3x cost (Futur eSearch).
Code/compute worker	Mid (Sonnet / Haiku 4.5 class)	Low– medium	Brief; sandbox	Return; artifacts	Haiku 4.5: 73.3% SWE-bench Verified, "similar... at one- third the cost" of Sonnet 4 (vendor, Anthropic). Tests are read-only to the worker (Imposs ibleBench).
Verifier	≥ worker tier by task,	High	Original brief; criteria[] ids +	Per- criterion	For correctness: a single judge at ≥

ROLE	TIER (DEFAULT)	EFFORT	MAY READ	MAY WRITE	NOTES
	family ≠ worker		evidence pointers; snapshots	verdicts	worker tier – judge accuracy tracks solver accuracy (JudgeBench : GPT-4o 56.6%, o1-preview 75.4%). For rubric/format criteria only: a three-judge cross-family small panel (κ 0.763 vs 0.627 for single GPT-4 on NQ, 7–8× cheaper – PoLL). The brief tags each criterion <code>correctness</code> or <code>rubric</code> .
Writer	Controller tier, fresh context	Medium	Ledger; artifacts by ref (mandatory for non- <code>checked</code> findings)	Draft	Single pass.
Grounding	Code + small entailment model	Low	Draft; snapshots	Claim labels	Entailment error rate is measured on a labelled set and stamped on every <code>entailed</code> label (\$7.4).
Deliverable gate	Verifier (correctness) or panel (rubric)	High	Report; <code>acceptance_criteria[]</code> ; Ledger objective	Verdict; one rewrite request	The report is the only artifact you see; it is verified like any Return.

Families and substitution. "Family" = vendor (Anthropic, OpenAI, Google are three). A Claude worker therefore requires a non-Anthropic Verifier. Rung 5 escalation must differ from the *worker's* family and be graded by a Verifier that differs from the *escalation model's* family; with three vendors this pins the assignment (e.g. worker A, Verifier B, escalation C graded by B or A-frontier — never C-by-C). If the required family is unavailable, the Rail pauses the rung and queues; it never silently falls back to same-family.

4. Depth tiers

Every task enters at the lowest tier that can answer it. Nova picks the entry tier — that is a model making a routing decision, and §14 notes it is unevaluated; mitigation is that the triggers below are Rail-checked and escalation is cheap relative to a wrong deep dive. Two well-known products' *lighter* tiers beat their own deep-research tiers on analyst tasks ([FutureSearch](#)), so the default is down, not up.

T0 — Direct. Nova answers from context. *Up-trigger*: the answer depends on a fact Nova would have to guess, or you ask for sources.

T1 — Lookup. One research worker, ≤ 10 tool calls, ≤ 3 minutes. Grounding on every factual claim. *Up-trigger*: `partial` / `blocked`, any load-bearing claim `unsupported`, or a comparison in the question.

T2 — Brief write-up. Scope (Nova, ≤ 2 clarifying questions, before any dispatch) $\rightarrow$ 2–4 disjoint research briefs in parallel, 10–15 calls each $\rightarrow$ Rung 1 $\rightarrow$ Verifier $\rightarrow$ compress $\rightarrow$ one Writer $\rightarrow$ grounding $\rightarrow$ deliverable gate. *Up-trigger*: >2 lines of inquiry `partial`, or a `conflicted` finding on a load-bearing criterion, or your request.

T3 — Deep dive. 1. *Scope*. Nova restates the question, writes `acceptance_criteria[]`, asks you ≤ 2 clarifying questions *now* (this is the cheap moment; later escalations are expensive). 2. *Pre-read*. Two mid-tier workers, 5 calls each, map the territory. Cost is $\sim 5\%$ of T3; it prevents locking a plan built on Nova's priors. 3. *Perspectives and outline, locked*. Nova writes the perspectives and outline from the pre-read. Outline-first raised organisation +25 points and breadth +10 on Wikipedia-style articles ([STORM](#)) — applied here by inference to reports generally. 4. *Read round*. 5–10 research workers, one per perspective $\times$ sub-question, ≤ 15 calls each. 5. *Compress*. After every round, Nova writes the findings ledger from Verifier-labelled Returns (one entry per finding: text, `refs[]`, label, perspectives served, `conflicts_with[]`). Forgetting earlier findings was the strongest failure correlate in Deep Research Bench traces (coefficient -0.843 , [FutureSearch](#)). 6. *Gap analysis and bounded replan*. Nova compares ledger to criteria; may revise the outline once here; spawns round 2 (and at most round 3) only for named gaps. A replan consumes a round. The Rail ends the loop when a round adds no novel evidence-bearing source *and* query diversity is exhausted (stall rules in §5). On short-answer search tasks 77–94% of episodes add nothing and a 20-episode cap retained 86–92% of accuracy ([2608.01913](#)); the long-form analogue is unmeasured (§14). 7. *Single write*. Writer from outline + ledger; opens artifacts by reference for every non-`checked` finding. 8. *Grounding and conflict pass*. Span-exists against snapshot $\rightarrow$ entailment $\rightarrow$ independence $\rightarrow$ labels. Conflicts are surfaced, never silently resolved. Unverifiable claims are labelled, never dropped, never promoted. 9. *Deliverable gate*. Report graded against `acceptance_criteria[]` and objective; one bounded rewrite; then to you with labels, conflicts, unmet criteria, and spend.

Budgets — see §9 for the worked model. Starting caps, in *cache-adjusted billed input-equivalent tokens*: $T1 \leq 150k$; $T2 \leq 1M$; $T3 \leq 5M$ — set above the typical-with-one-escalation figures in §9 so a single Rung 5 does not breach the cap. Wall-clock targets: $T1 \leq 3$ min; $T2 \leq 12$ min; $T3 \leq 45$ min. On budget stop the Rail forces partial Returns from running workers, Nova compresses what exists, the Writer runs on the partial ledger with every finding from an unverified round labelled `unsupported`, and the deliverable states which criteria are unmet.

5. Drift controls

Ranked by measured effect; grade in brackets.

Goal drift → **the objective travels verbatim and is criterion zero.** A goal-restating scaffold kept Claude 3.5 Sonnet near-perfect past 100k tokens where every unscaffolded model drifted ([Arike et al., eval repo](#) — unreviewed, one model family; C). Field 2 of every brief is the Ledger `objective`, which Nova writes *in third person* at scope time and you confirm; it is then immutable and string-matched by the gate (so your phrasing never reaches workers and the match is exact).

Plan drift → **pre-read, then lock, then one bounded replan.** Plan-then-execute cut tokens 64% and gained +4.4 points over step-wise reacting across six benchmarks ([ReW OO](#), 2023 models; D as applied to research outlines). Coding agents that reach the right code then overwrite it account for 32–40% of edit-quality failures (~23% of all failures); five intermediate patches bit-identical to the correct one were discarded and all five were recoverable with an edit-commit checkpoint ([TRAJEVAL, 2026](#)). Every worker artifact is a committed snapshot; nothing is overwritten in place.

Scope drift → **the Verifier grades against the brief; the deliverable gate grades the report against the criteria.** +15.6% from a verification step checking output against the high-level spec ([MAST](#); B).

Semantic drift across hops → **one structured hop per direction; artifacts by reference; the ledger treated as a lossy hop.** Compaction loses artifact facts (paths, what changed) worst of all dimensions — 2.19–2.45 out of 5 across three production compactors ([Factory](#); B). Hence the Writer must open artifacts for anything not `checked`, and ledger entries carry `refs[]` that the grounding pass checks against, not Nova's text.

Instruction drift in Nova's own long conversation → **Nova works from the Ledger, and the Rail compacts Nova's transcript.** System-prompt adherence drifts significantly within eight rounds ([Li et al.](#)); instruction-following fell 0.877 → 0.707 by turn 3 ([Multi-I F](#)). The Ledger, not the transcript, is re-injected at each decision; §7.5 gives the compaction policy.

Sycophantic drift → **no worker sees your words; the Verifier sees pointers, not prose.** Judges flip 84.5% under a bare rebuttal ([Kim et al.](#)); third-person framing cut sycophantic flips by up to 63.8% in a debate setting ([SYCON-Bench](#)). The Verifier's inputs are exactly: the brief, `criteria[].{id, evidence_pointer}`, and the snapshots those pointers resolve to. It does not receive `summary`, `claims[]` text, `assumptions_made`, `failure.message`, or `blocked_on` — those go to Nova *after* the verdict.

Stall → **Rail-owned, on repetition and novelty, defined precisely.** A worker is stopped (with a forced partial Return, not a kill) when any of: same action+observation 4×, same action+error 3×, alternating ping-pong 6× ([OpenHands thresholds](#)); or two consecutive rounds with zero *novel sources*, where a novel source is a normalised URL whose content hash is not already in the snapshot store. A T3 gap loop ends when a round adds zero

novel sources across all workers *and* Nova's next-round query set overlaps $\geq 80\%$ (normalised) with prior queries — both conditions, so sparse topics with bad queries get one more chance with new queries rather than a confident thin report. Redundant re-querying is "the cleanest behavioral predictor of failure" in research traces ([2608.01913](#)). The worker-side `stop_hint` in the brief is advisory only.

6. The canonical brief

Final form. Delimited sections; XML tags for Claude-family workers, Markdown headings for OpenAI-family; structure matters, syntax does not for frontier models ([He et al.](#); [2602.05447](#)). Under 2k tokens excluding inputs; ≤ 5 hard constraints.

Section order is chosen for cache stability first, then for the long-input rule: the sections that are identical across many briefs (identity schema, policy, output schema) sit first so the prefix caches; large variable inputs sit last, immediately followed by a one-line restatement of the objective and criteria ids (the "instruction adjacent to data" rule — local instruction repetition lifted GPT-4o from 31.5% to 54.3% at 16k tokens, [LongIns](#)). Cached input is $\sim 0.1 \times$ list price ([Anthropic pricing](#)).

```
<brief>
<missing_information_policy>          ← constant text, cached
  If an input needed for any criterion is absent or ambiguous: return status=blocked with the
  precise question and candidate answers. If a reasonable assumption exists and the criterion is
  not marked load_bearing: proceed and record it in assumptions_made. Never fill a gap silently.
  If the task is infeasible as specified, or inputs contradict the criteria: return status=rejected
  with both sides. Reporting a criterion unmet is acceptable; claiming it met without the named
  evidence is not.
</missing_information_policy>

<output_schema>                      ← constant per role, cached (§7.2)
</output_schema>

<identity>
  task_id · parent_id · attempt N of MAX · tier · role
  if attempt > 1: previous_failure_class · criteria the Verifier marked unmet, verbatim
</identity>

<objective>
  [Ledger objective, verbatim, third person. "The task is to return ..."]
</objective>

<scope>
  in: [...]  out: [...; named sibling tasks not to duplicate]
</scope>

<constraints>
  [ $\leq 5$ , positive, content → format, precedence notes for any pair that could conflict]
</constraints>

<budget>
  max_tool_calls · max_turns · max_wall_time
  stop_hint (advisory): a round with no source you have not already cited is a signal to finish
</budget>

<definition_of_done>
```



```

C0 (always): the output serves the objective above.
C1 [correctness|rubric] [load_bearing?]: [the observation that proves it]
C2 ...
</definition_of_done>

<sources> (research/extraction only)
  allowed tiers · do-not-cite: user-generated content unless corroborated by a non-UGC source
  rule: every factual claim carries ref_id + quoted span; fetch only through the Rail proxy
</sources>

<sandbox> (code only)
  writable paths · test command (run-only; tests are read-only) · abort: if tests and spec conflict,
  return rejected with both
</sandbox>

<inputs>                                     ← large, variable, last
  [verbatim records/documents; everything else as ref_id · title · one-line extract]
</inputs>

<restate>
  Objective: [one line]. Criteria: C0–Cn. Output per <output_schema>. Reasoning, if any, in <working>
  before the block.
</restate>
</brief>

```

Wording rules (prior report): positive constraints, no emphasis markers, no persona line, no motivational framing, two format-faithful examples for extraction workers in `<output_schema>`.

7. Ledger, Return, verdict, labels, and Nova's memory

7.1 Task Ledger (Nova-owned, Rail-enforced, versioned)

`objective` (immutable after scope) · `acceptance_criteria[]` (id, text, type `correctness|rubric`, `load_bearing`, observation) · `tier` · `budget` / `spent` (billed-equivalent tokens, wall time, per-rung attribution) · `plan` + `plan_version` · `findings[]` · `open_gaps[]` · `assumptions[]` (with Nova's accept/reject) · `conflicts[]` · `decisions[]` (timestamp, reason, model id, prompt version, brief version) · `escalation` · `follow_ups[]` (a later question inherits this Ledger at T1 and escalates normally).

7.2 Return (worker → Rail)

- `criteria[]` — {id, met, evidence_pointer}; the pointer is a snapshot id + locator (file/line, URL/hash/span offsets, command + captured stdout hash).
- `artifacts[]` — {id, type, snapshot_hash}.
- `status` — `done` | `partial` | `blocked` | `rejected` | `failed`.
- `blocked_on`.
- `assumptions_made[]`.
- `failure` — {class, retryable, message, attempted[]}.
- `side_effects[]`.
- `usage` — calls, tokens, wall, `novel_sources_this_round`, verbalised confidence (triage only).
- `claims[]` — {text, ref_id, span_locator}.
- `summary` ≤150 words.

Items 1 and 2 go to the Verifier (pointers and snapshots only). Items 3–10, including `usage` and its verbalised confidence, go to Nova after the verdict.

7.3 Verifier verdict

Per criterion `{id, verdict: met | unmet | unverifiable, checked: [pointers], reason}`; `C0 serves_objective` graded from artifacts, not from any description of them; `conflicts_detected[]`.

7.4 Conviction labels — named after the check

LABEL	EARNED BY	WHO ASSIGNS	ERROR RATE CARRIED
<code>checked</code>	A deterministic check passed against a snapshot: test executed in a fresh sandbox with stdout hash match; schema validated; span-exists at locator; artifact hash matches.	Rail	~0 (the check is exact)
<code>entailed</code>	<code>checked</code> span-exists <i>and</i> the entailment model says the span supports the claim.	Rail + small model	The entailment model's measured false-positive/negative rate on your labelled set, stamped on the label (e.g. <code>entailed@0.08</code>).
<code>corroborated</code>	<code>entailed</code> by ≥2 sources that pass the independence heuristic (different registrable domain; no shared byline/wire marker; content similarity below threshold).	Rail	Heuristic misses syndication; treated as <code>entailed</code> -grade for load-bearing decisions until the heuristic is evaluated.
<code>sample-agreed</code>	Two small-model extraction samples agreed at field level (normalised). Not <code>checked</code> . Correlated samples agree on the same wrong answer; this label gates escalation, it does not prove correctness.	Rail	Unknown until measured (§13 Rung 2 precision); until then it carries no evidential weight for load-bearing decisions – a load-bearing extraction field must additionally be <code>checked</code> (schema + source span) before the Writer may rely on it.
<code>single-source</code>	One <code>entailed</code> source.	Rail	as <code>entailed</code>
<code>unsupported</code>	Asserted with no span, or span does not entail. Kept, labelled, never promoted. Synthesis claims that rest only on other labelled claims are <code>derived(from: ids)</code> , not <code>unsupported</code> .	Rail	–
<code>conflicted</code>	Independent <code>entailed</code> sources disagree. Surfaced with both sides. Nova may downgrade one side only on a <i>mechanical</i> ground (source is UGC; source is superseded by a later version of the same publisher's document) and must log it; Nova never resolves a conflict by preference.	Rail / Nova (logged)	–

When Nova merges findings into a ledger entry, the entry takes the *weakest* label among its sources unless every source is independently `entailed`, in which case `corroborated`.

7.5 Nova's own context

The Rail compacts Nova's transcript whenever it exceeds a threshold (start: 60% of window), using a fixed procedure: the Ledger is never compacted (it is re-injected whole); the transcript is replaced by `decisions[]` plus the last two turns; artifact facts are never summarised into prose, only referenced. In T3 the ledger is pruned for re-injection to entries touching `open_gaps[]` and load-bearing criteria, with the rest available by reference. This is the single largest drift surface the prior design ignored.

8. Conviction and efficiency ladder, in cost order

Rung 0 — Pre-send gate (code, plus a ~20–90 ms classifier; bounded at 3 Nova ↔ gate bounces, then escalate to you). Checks: required sections present; `objective` string-matches the Ledger after whitespace normalisation; ≤5 constraints; no known-impossible constraint pairs; every criterion names an observation of *an allowed observation type* (command+expected output, file/field, URL/span, schema) — free-text "observations" bounce; every criterion tagged `correctness|rubric`; token count within tier; every `ref_id` resolves; injection scan on embedded third-party text ([Prompt Guard 2](#): 88.7–97.5% recall at 1% FPR). Specification failures are ~44% of multi-agent failures ([MAST](#)); constraint count alone predicts compliance within ~10% ([2509.21051](#)). This gate checks *form*; a well-formed vague brief passes, and the bounce-reason metric (§12) is how you find out.

Post-dispatch injection. Workers fetch through a Rail proxy that snapshots and scans every fetched document before it enters the worker's context; artifacts are untrusted until scanned. The brief-boundary scan alone protects nothing the worker reads later.

Rung 1 — Snapshot re-check (code; not free in time, but no side effects). Every `evidence_pointer` resolves against the content-addressed snapshot captured at worker time — never a live re-fetch. Commands are re-executed only if declared side-effect-free in the brief and then in a fresh sandbox; otherwise the check is "captured stdout hash matches." Failure → `unmet` regardless of the worker's claim. This is the strongest conviction mechanism available: it is what makes sample coverage cashable (SWE-bench Lite 15.9% → 56% at 250 samples *with* a verifier, [Large Language Monkeys](#)). Many research criteria are not mechanically checkable; those are tagged `rubric` at scope time and skip to Rung 4.

Rung 2 — Agreement gate, small extraction workers only (2–3× small-model cost). Two samples; field-level comparison after normalisation (dates, units, whitespace, case) with a per-field tolerance declared in the schema. Agree →

sample-agreed, continue. Disagree on any load-bearing field → third sample; 2-of-3 wins, else escalate the same brief once to the **mid tier** (this is the extraction escalation path; Rung 5 is the Verifier-disagreement path, and they never chain). Five-sample agreement raised failure-prediction AUROC 54.8% → 92.7% on arithmetic ([Xiong](#)) but is weak on knowledge tasks and unreliable for frontier models ([2607.08065](#)); that is why it gates escalation and never earns `checked`.

Rung 3 — Grounding (one small-model call per claim). Span-exists (snapshot) → entailment → independence → label. Vendor-reported: span-level citation enforcement took one customer's source hallucinations from 10% to 0% ([Anthropic Citations](#); C). Search-augmented claim verification agreed with humans 72% and won 76% of disputes at >20× lower cost ([SAFE](#)) — which also means 28% disagreement, hence the stamped error rate. Scale of the problem: even GPT-5 averaged 43.8 unsupported claims per market-analysis report and Open Deep Research 91.9 ([LiveResearchBench](#)).

Rung 4 — Cross-family Verifier (one call at high effort; ≈ cost of the worker run). Inputs per \$5. Correctness criteria: single judge at ≥ worker tier by task; rubric criteria: three-judge small panel. Expect ~2 effective independent votes however many judges you add (9 judges across 7 families gave $n_{\text{eff}} \approx 2.18$, [2605.29800](#)); never buy more than three. Verification costing as much as generation is the steady state of this design, not a bug.

Rung 5 — Disagreement-triggered escalation (one frontier call + re-grade). Trigger: the Verifier marks a *load-bearing* criterion `unmet` that the worker claimed `met`. Non-load-bearing disagreements go to Nova as `unsupported`. The same brief goes once to a frontier model from a family ≠ worker; the Verifier (family ≠ escalation model) re-grades. A two-model cascade captured up to 79.5% cost reduction at 90% of ceiling quality and fixed chains deeper than two underperformed the pairwise envelope ([2605.06350](#)). Never a third stage; after Rung 5 the result is labelled as it stands.

Rung 6 — Nova's judgement. Only for criteria marked `unverifiable`. Reference-guided: Nova sketches the expected answer first, then grades (cut GPT-4's math-judging failure rate from 70% to 15% in [MT-Bench](#)); Nova never sees the worker's argument.

Rung 7 — You. The question, the candidates, a deadline, and a declared default. **Default is park** for anything with declared `side_effects[]`, any load-bearing criterion, or any T3 run; "proceed on stated assumption, labelled `unsupported`" is allowed only for non-load-bearing claims. If you answer after a default fired, the Rail re-opens the task at the affected criterion and logs the rollback.

Efficiency rules across the rungs. Static role-to-tier assignment: learned routers recover only 7.5–14.4% of oracle gains ([2608.08265](#)) and the top fifteen differ by 0.23 points ([Routing Plateau](#)) — evidence that one alternative is bad, not that static is optimal (B). Effort low for workers, high for Verifier and for math-like verifiable steps ([FutureSearch](#)); model tier beats token budget: "upgrading to Claude Sonnet 4 is a larger performance gain than doubling the token budget" ([Anthropic](#)). Retriever quality before iteration count: a better retriever was worth ~15 points on BrowseComp-Plus and reduced calls ([2508.06600](#)). Tool results capped per call (start: 4k tokens, extractive) so worker context stays bounded. Concurrency caps per provider and per search API, with backoff, set in the Rail.

9. Worked cost model

Assumptions: tool results capped at ~4k tokens; research worker context grows to ~30–40k by call 10–15; cache reads at 0.1×; "tokens" throughout means *billed input-equivalent* (cache-adjusted input + output). Figures are order-of-magnitude and must be replaced by your own logs.

TIER	WORKERS	VERIFIER	NOVA (SCOPE, COMPRESS, DECIDE)	WRITER	GROUNDING	TYPICAL TOTAL	WITH ONE RUNG 5
T1	1 × ~40k	1 × ~15k	~10k	—	~10 claims × 1.5k = 15k	~80k	~130k
T2	3 × ~120k = 360k	3 × ~40k = 120k	~60k (2 compress + decisions)	~25k	~60 claims → 90k	~650k	~900k
T3	R1 7 × ~150k + R2/R3 ~6 × 150k ≈ 2M	~15 × 40k = 600k	pre-read 2 × 30k; compress ×3 ~150k; decisions ~40 × pruned ledger ~15k = 600k	~60k	~300 claims → 450k	~3.9M	~4.5M

Where the ladder multiplies: Rung 2 makes extraction 2–3× by construction (4–6× if tolerances are too strict); Rung 4 is ≈1× the worker again; Rung 5 adds a frontier run of the same brief plus a re-grade — in the table that is +50k at T1, +250k at T2, +600k at T3, i.e. roughly 1.3–4× the escalated worker's own cost depending on tier — and, if the false-success baseline holds, will fire on a meaningful minority of Returns unless the load-bearing filter is used; ledger re-injection in T3 is the quiet quadratic and is why \$7.5 prunes. Wall-clock for T3 is three sequential rounds each bounded by the slowest worker plus a high-effort Verifier pass: 30–45 minutes is realistic.

The honest summary: T2 costs roughly 2–3× a single-agent run of the same task and T3 roughly 3×; the premium buys labelled claims, a verified deliverable, and bounded drift. If you want cheaper, drop Rung 4 for rubric criteria and Rung 2 for non-load-bearing fields first; never drop Rung 1 or the deliverable gate.

10. Evidence map

Grades: **A** replicated benchmark/controlled study; **B** single benchmark or preprint, or instrumented production report; **C** vendor guidance or practitioner consensus; **D** inference from adjacent evidence.

DECISION	SOURCE	GRADE
Single controller; Rail owns bounds	Google/MIT 4.4× vs 17.2× (one preprint); MAST +9.4/+15.6% (one framework)	B
Workers read, never coordinate; parallel reads only	Cognition; LangChain ODR disjoint writing; Anthropic research system	C, converging
Deterministic Rail carries handoffs; every loop bounded	OpenAI "via your code"; Stripe blueprints; Hou et al. 68 loop defects	B/C
Fixed-key briefs and Returns; no transcripts	Laban (multi-turn chat, by inference); relay study (weak-relay condition)	B/D
Ledger treated as a lossy hop; Writer opens artifacts	Factory compaction 2.19–2.45/5 on artifacts; relay persistence 83–100%	B
Objective verbatim, third person, criterion zero	Arike eval repo (unreviewed); Anthropic contract; SYCON third-person	C/B
Pre-read → plan lock → one replan	ReW00 (2023, by inference); TRAJEVAL checkpoint recovery (coding)	D
Verifier grades against brief; deliverable gate grades report	MAST +15.6%	B
Verifier family ≠ worker; ≥ worker tier by task; panel for rubric only	Panickssery; JudgeBench; PoLL; n_eff 2.18	A
Verifier sees pointers + snapshots, never worker prose	Confident Closing AUROC ≤0.65 on closing language; Kim et al. 84.5%	B
Snapshot re-check before any model judgement	Large Language Monkeys (verifier-limited selection); ImpossibleBench (editable checks get gamed)	B/D
Agreement gate = escalation trigger only, small workers only	Xiong M=5; 2607.08065	B (scope-limited)
Span grounding + entailment with stamped error rate	SAFE 72%/76%; Anthropic Citations (vendor anecdote); LiveResearchBench	B/C
Labels named after checks; no scalar	Xiong ECE; 2607.08065; 2604.23505 (framework paper)	B

DECISION	SOURCE	GRADE
Selection over synthesis	Selection bottleneck 0.81 vs 0.18	B
No debate	Smit et al.; Zhang et al.	A
No blended MoA	Selection bottleneck 0.18. Note: a 2026 compute-matched study (2605.01566) finds MoA <i>beats</i> self-consistency by +2.7 points at equal compute and +7 at 20x – MoA is excluded here because blending loses to selection, not because it fails to beat sampling	B
Static tier assignment	LLMRouterBench/Routing Plateau/2608.08265 (routers bad ≠ static optimal)	B
Small models for short-doc extraction, mid for long	ExtractBench (Gemini 3.5 Flash)	B
Effort low for research, high for verification	FutureSearch effort paradox (research); AIME effort curves from third-party harness, not the cited page	B/C
Two-stage cascade maximum	2605.06350	B
Gap loop bounded by novelty + query-diversity	2608.01913 (short-answer tasks)	D for long-form
Outline before retrieval	STORM (Wikipedia-style)	B for that genre, D generally
Findings ledger	FutureSearch forgetting coefficient; Tongyi IterResearch; Anthropic harness	B/C
Single writer from labelled evidence	LangChain ODR; LiveResearchBench aggregation	B
Unverifiable labelled, never dropped	Claude Code /deep-research; Tow Center >60% citation failure baseline	C/B
UGC down-weight	WARP 17–23% UGC; 38–51% mention rate from a single poisoned URL	B

DECISION	SOURCE	GRADE
Stall rules	OpenHands; Magentic-One ≤ 2 ; 2608.01913 redundancy predictor	B/C
Human escalation: park default, deadline, rollback	Temporal durable timers; ImpossibleBench abort effect	C
Pre-send gate	MAST spec failures; 2509.21051; Stacking Collapse conflicts; SWE-bench Verified screening (benchmark tasks, not briefs)	B/C
Tier ladder, default down	FutureSearch; Anthropic effort rules; Gemini two-tier	B/C

11. Deliberately excluded

Agent-to-agent messaging (MAST chatty frameworks fail 41–87%); multi-agent debate ($\approx$ self-consistency at higher cost — [Smit](#), [Zhang](#)); blended MoA synthesis (loses to selection; see the honest note in §10); learned per-query routing (plateaued); persona lines, emphasis markers, motivational framing, "don't do X" constraints (null or negative); reflection loops without an external signal (flat or negative); worker write access to tests, shared state, or other workers' artifacts (30–76% reward hacking when the checker is editable); any unbounded loop; any live re-fetch as a verification step.

12. Production requirements the design now states

- **Permissions and secrets.** Per role: network allow-list (workers only via the Rail proxy), filesystem paths, credentials scoped per task and revoked at Return; the Verifier has no write access anywhere; sandboxes are fresh per attempt and destroyed after snapshot.
- **Observability.** One event log keyed `run_id → task_id → attempt → rung`, with model id, prompt/brief/template version, tokens, wall, cost, and label transitions; full replay from snapshots. §13's metrics are computed from this log.
- **Versioning.** Brief templates, the constant brief sections, role → tier table, entailment model, and thresholds are versioned; every `decisions[]` entry records the versions in force.
- **Partial results.** On budget or stall stop: forced partial Returns; unverified-round findings labelled `unsupported`; deliverable lists unmet criteria and spend.

- **Progress and cancellation.** Per-tier progress events (stage, round, spend vs budget, findings so far); cancellation at any Rail checkpoint returns what exists.
 - **Provider outage.** Required-family unavailable → rung paused and queued with a deadline; never same-family fallback; you are told.
 - **Evaluation harness.** A labelled set of ~50 tasks per tier with ground truth and human-graded criteria, used to measure false-success rate, entailment error rate, and agreement-gate precision before thresholds are trusted — the design's numbers are literature priors, not yours.
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13. Instrumentation

Per run, from the event log: run-level best/worst spread on repeated tasks (Vending-Bench: every model has derailing runs while means look fine — [Andon Labs](#)); false-success rate (`done` with any `unmet` ; literature 13–79%); `unsupported` claims per deliverable (best systems ~44–92 per report); redundant-query ratio per worker; pre-send bounce rate *by reason* (high "criterion not observable" = Nova's brief-writing is the problem); tokens per `entailed` -or-better claim by tier; Rung 5 trigger rate and human-escalation rate; `conflicted` labels per deliverable (zero is suspicious); entailment false-positive/negative on the labelled set; Rung 2 precision (sample-agreed but later `unmet`).

14. Open questions

1. Most load-bearing numbers are 2024–25 vintage; the one 2026 replication (sample agreement) showed the signal weakening. Re-measure on your tasks before tuning.
2. Recitation, progress ledgers, reference-not-summary, deterministic rails: universally shipped, never ablated (grade C).
3. Gap-loop saturation for long-form work is unmeasured; the novelty + query-diversity rule is a design choice.
4. Whether the `rejected` /abort path reduces gaming on Claude-family workers (ImpossibleBench's 6× effect barely moved Opus 4.1).
5. Effective independence of cross-family checks on long-form outputs; ~2 effective votes is measured for judges on classification, not reports.
6. Style-consistency cost of crossing families at the Verifier/escalation boundary: unmeasured; mixing is confined to selection points.
7. Nova's entry-tier choice is a model routing decision with no evaluation; the default-down rule and cheap up-triggers are the mitigation.

8. A source-credibility scorer: universally needed, not published; the UGC rule is a placeholder.
9. The independence heuristic for **corroborated** will miss syndication until evaluated.
10. If you later need a guarantee (" $\leq 10\%$ of **corroborated** wrong"), a conformal wrapper on the Verifier's acceptance ([SCOPE](#), pairwise judging) is the evidenced route and needs your labelled set.